DSA0504 – QUERY PROCESSSING AND DATA SCIENCE

LAB EXPERIMENTS

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Experiment 11

Aim

To create a DataFrame with ten rows and four columns containing random values, convert some values into **NaN**, and highlight the NaN values.

Algorithm

1. Import the Pandas and NumPy libraries.
2. Generate a DataFrame with random values.
3. Replace some values with NaN.
4. Define a function to highlight NaN values.
5. Apply the style to the DataFrame.
6. Display the styled DataFrame.

Python Code

```
import pandas as pd
import numpy as np

df = pd.DataFrame(
    np.random.randn(10,4),
    columns=["A","B","C","D"]
)

# Convert some values to NaN
df.iloc[2,1] = np.nan
df.iloc[5,3] = np.nan
df.iloc[8,0] = np.nan
```

```
# Highlight NaN values
styled_df = df.style.highlight_null(color="yellow")

print(df)

styled_df
```

Input

Random DataFrame (10 × 4)

Output

	A	B	C	D
0	0.54	-0.12	0.88	1.32
1	-1.11	0.65	0.45	-0.76
2	0.98	NaN	1.02	0.25
3	-0.63	0.71	-1.22	0.81

...

```
>>> ==============================================
            A         B         C         D
0 -1.086840  0.449443 -1.921366  0.688359
1  1.920889 -1.609185  0.247128  2.044193
2 -0.054901       NaN -0.978931 -1.326821
3  0.896442  2.479553  0.159078  0.383096
4 -0.779953 -0.811010 -0.707561 -2.082866
5 -0.503075 -0.032926 -0.243877       NaN
6  0.161637 -1.111016  0.686368  0.228618
7 -0.933600  1.007008 -0.966028 -0.548169
8       NaN  0.043211  0.909473  0.602303
9  1.244963  0.178217 -0.160390  0.388838
>>>
```

Result

Thus, the DataFrame was successfully created, NaN values were inserted, and highlighted using Pandas Styler.

Experiment 12

Aim

To create a DataFrame with ten rows and four columns and set the **background color to black** and **font color to yellow**.

Algorithm

1. Import Pandas and NumPy.
2. Generate a random DataFrame.
3. Apply black background and yellow font color using Pandas Styler.
4. Display the styled DataFrame.

Python Code

```
import pandas as pd
import numpy as np

df = pd.DataFrame(
    np.random.randn(10,4),
    columns=["A","B","C","D"]
)

styled_df = df.style.set_properties(
    **{
        "background-color":"black",
        "color":"yellow"
    }
)

print(df)

styled_df
```

Input

Random DataFrame (10 × 4)

Output

	A	B	C	D
0	0.24	-0.78	1.55	-0.36
1	1.02	0.11	-0.47	0.88

...

Displayed with:

- Background → **Black**
- Font → **Yellow**

```
>>
= RESTART: C:\Users\senthil\AppData\Local\Progra
          A         B         C         D
0  1.111766 -0.228996 -0.251951 -0.064030
1  0.479152 -0.196994  0.116267 -1.139769
2 -0.992372 -1.252836  0.099058  0.165117
3  2.256681 -1.156815 -0.187937  1.391768
4 -1.538034  0.798827  0.621802  2.624677
5  1.360865 -1.055341  0.383531 -1.069924
6  0.522299  0.729582  0.373623  0.537920
7 -0.718115 -0.639296 -0.074706  0.609369
8  0.007479 -0.324241  0.929770  0.213702
9  0.494997  0.244571  0.718106 -0.350679
>>
```

Result

Thus, the DataFrame was successfully displayed with a black background and yellow font using Pandas Styler.